

Video Transcripts

Module 4: Designing Artificial Machines to Solve Problems

Video 1: Implementation and Design of AI (03:08)

In the next two modules, we have two key objectives. The first is to review the four-stage AI design process to create a product that is built entirely with AI technologies. The second objective is to do a deep dive on a specific AI technology called transformers. Transformers are the key technology behind large language models available commercially.

The first one of its kind to hit the market broadly was ChatGPT. Ever since, many others have joined, broadening the scope into multimodal generative AI offerings that even include sound, image, and video. To put in perspective what LLMs mean for a research institution like MIT. Note that from a scholarly point of view, the paper that introduced the technology, whose title is "Attention Is All You Need", is well on its way to becoming the most-cited research paper of all times. That's a bit mind blowing for a simple researcher like me, but it's accumulating citations at the highest level that I know of.

In this module, we will design a service to discriminate whether a person has the onset of a given disease or not. In the next module, we will present the first two stages of a transformer, the tokenizer and the encoder, and discuss how the output of the decoder can already be used to improve many discrimination problems. This is especially true when there is text that can add information to the task at hand such as discriminating if someone has dementia or not.

We will first present an AI model that determines if someone has dementia or not by analyzing the speech and then explain how the first two stages of an LLM can process natural language to improve the discrimination accuracy.

We continue in the next module with a deep dive on the third stage of a transformer, the decoder. Here is where we will try to understand the magic of ChatGPT, and how it creates such as astonishing human-like results. We will also aim to understand some of the limitations of generative AI technologies. This deep dive on transformers will help you evaluate when ChatGPT

and other LLMs can be useful for a business problem, and how their shortcomings may be taken into account to achieve real business value.

We will also briefly discuss the pending hurdles to develop a true general intelligence machine that can behave autonomously like a human being, including a brief review on some approaches being explored at MIT. The transformer architecture has three distinctive components that operate sequentially, tokenizer, encoder, and decoder. If there is one thing that I would like you to remember from the generative AI module, it is a basic understanding of what each of these three stages does.

Video 2: Stage I: Figure–Ground and AI Categorization (06:38)

We will look at the different steps to design an AI product that listens to your voice and tells you whether you have a risk of a certain medical condition like Alzheimer's, Dementia, or COVID. To complete the process, one can answer the eight questions we described first, really in any order. For pedagogical reasons, we'll follow the same order that we did. The first step then in our journey is to identify the superhuman intelligence we want to achieve. No one has issues identifying the image on the screen as containing a raised cabin. This, despite it being occluded by a bunch of vegetation. Scientists and engineers talk about segregating figure from ground as a first step in human visual recognition and also in computer vision. In this particular case, it can be done sort of unambiguously.

Now, look at this other image. In these other cases, the separation between figure and ground may be ambiguous like here where you can think of the ground as black, and the figure be a white base or conversely you can think of the ground as white and the figure as two black silhouettes looking at each other. The separation of figure and ground also occurs in sound. In this short audio clip, you can hear a conversation between two people with background noise. You can distinctively recognize what they're saying and at the same time identify the background noise telling you the context is a cocktail party or on a street with traffic. Scientists believe there are some similarities between figure-ground segregation in the first example with computer vision in the cabin or what occurs here with sound where you're discriminating sound texture from the real sound you want to hear.

As another audio example, if you are in a car driving at 50 miles per hour and a window is slightly open, chances are you will notice it through the sound made by the air flowing into the car. This will happen independently of the make of the car, the amount of people in the car, whether you're listening to the radio or not, having a conversation with passengers, or talking onto the phone. The human ear can segment sounds very much like you can differentiate the forest from

the cabin. Mechanics often listen to odd engine noises to diagnose if there may be an issue with the vehicle. Common engine problems that the human ear can be trained to diagnose include noises from the valve train, from the timing chain, from pre-ignition, from the crankshaft bearing, from the piston's lab or pin, etcetera.

Now listen carefully to the sound of this engine. And now to the sound of this other one. Can you hear any significant differences? Both correspond to the exact same type of vehicle, however one of them has the air filter in poor condition and the other one does not. Let me give you another chance. The first one again. And now the second one again. Unless you are extraordinarily gifted, you won't notice the difference. Both sounds are recorded with a simple phone. And what's pretty amazing is that AI can be taught to listen to the differences between the sound with the air filter incorrect and the sound with the new filter. And training is done with more careful data gathering techniques so that the AI algorithms don't have to deal with figure-ground issues which is helpful because that way one needs a smaller sample. Researchers in my laboratory, the Auto-ID laboratory, did some tests to see if they could train AI to distinguish the sounds and found it was indeed possible.

The image you're seeing shows Fourier transform features illustrating how they vary depending on how many dirty layers are incorporated in the filters so that they cannot only tell you whether it's dirty or not but how dirty it is and maybe even give you a forecast as to when you have to change the air filter. This means your phone, via its microphone, can extract information related to your air filter and provide you with an indication as to whether you have to change it or not. There is a whole field growing on diagnosis using the phone and not using the microphone only.

In another related piece of engineering research, the same team of researchers in my lab found wheel imbalance can be detected using the accelerometer on the phone just with one phone. They demonstrated that in two classes of vehicles, in 90% of the cases they could detect whether there was an imbalance or not. Wheel imbalance has a lot of negative effects on comfort, safety, and durability of mechanical parts. But hopefully soon your phone will tell you when to go to the mechanic and why. It may even pre-order the pieces you need or the service that your vehicle requires. Several startups and vehicle manufacturers are entering this space. The overall idea is to have sort of a Shazam for vehicle sounds where you aim the phone at the car, and it tells you the type of sound it's listening and what are the issues. Is it doing well? Is it not doing well? Do you have to change this part or that other part?

Video 3: Stage I: Figure–Ground and AI Categorization — Digital Health (04:05)

Now going back to our health ambitions, we would like to do, essentially, the same done with vehicle diagnostics in the examples from the previous videos but with medical conditions in humans. Sounds had been used in medical diagnosis by humans for a very long time. This image features Laennec, the inventor of the original stethoscope, in 1816, examining a patient with probable tuberculosis. He was a skilled musician and physician whose art in building his own flutes led him to build the first stethoscope. He quickly became medical director at a clinic, and unfortunately, died shortly after of tuberculosis himself. The disease he helped diagnose so well.

Many other similar instruments were built shortly after, as you can see, on the image. These instruments served to diagnose certain conditions like regurgitation of the human heart and many others. And there are many conditions that can be detected just through changes in the voice even without any device. For example, Air cysts, which although usually asymptomatic, may accumulate respiratory secretions that become infected and lead to coughing or tracheobronchitis. Using a device, you can detect vocal fold polyps, which are usually benign and disappear in less than eight months, but that may require surgery on occasions. Bronchopulmonary dysplasia, or chronic lung disease, is a more serious condition that also influences voice, and that can be detected just with audio.

Until AI came along, the use of these instruments was a bit expensive for massive screening, not because of the instrument, but because of the intensive use of qualified personnel. As an editorial of the New England Journal of Medicine put it not too long ago, the tools were slowly coming of age because doctors would have incentives to pursue a more definitive diagnosis using more invasive techniques like MRIs. In fact, many doctors have not even exposed today to proper training on audio diagnostics.

There are many more conditions where the vocal cords may provide hints, as shown on the screen, including Chronic Otitis, Multiple Sclerosis, Parkinson's, or Hypothyroidism. Perhaps AI can play role in all of this. That technology should be superhuman should not be new.

In fact, humans can sense fever, but thermometers can do much better jobs, since they're objective, repeatable, and can provide a sort of finer scale than humans which can assess the evolution of temperature over time, or the surpassing of certain thresholds which require hospitalization. The difference with a thermometer is that AI is a generic technology that can do a

lot more than just give us a metric. One area that's a bit controversial is the use of voice processing to diagnose mental disorders. In one recent experiment, they found that automated analysis of free speech predicts psychosis onset in high-risk youths before doctors can even detect it. In the example, we are trying to address, we will see superhuman behavior of AI very much like in this psychosis example.

Video 4: Stage I: Figure–Ground and AI Categorization Alzheimer's — Example (05:20)

Take as another example, Alzheimer's, which represents about 75% of all dementias and is said to grow in the next decades given the expanding life expectancy. About 50% of those over 85 years old already have it. Doctors use different tests to decide if somebody has Alzheimer's or not because there is really no objective test that can do it. One of the tests they use is to have patients describe the image that you are seeing, the so-called Cookie Theft picture from the Boston Diagnostic Aphasia Examination. There are many other tasks and systems to assess Alzheimer's. Some of them involve repeating words or their spellings, etcetera. The reason why this picture is useful is because it can help identify some common cognitive-linguistic impairments of Alzheimer's patients when describing it. If one or more of the impairments are clearly present, the person has some deficiency that may indicate possible dementia.

Some examples of this shortcomings may include perceptual impairments. They won't realize, for example, that there are three subjects in the picture. As a second example, poor reference. So, the person may say they're cooking. It isn't clear who they refers to and, in fact, they aren't cooking really. And as a third example with poor inferences, they may say “the lady” instead of the “mother” has water running out of her instead of overflowing from the sink. Here, AI could use and has been demonstrated already, natural language processing techniques, to identify the various possible cognitive decay indicators to establish a dementia scale with the aim of identifying onset and progression of the disease with possibly, which hasn't yet achieved, finer metrics associated to each dementia type.

So, in terms of the two decisions to be identified in the first stage, we can state the first one as being able to discriminate between Alzheimer's positive and negative subjects with an accuracy of over 99%, and with a scale that tracks disease progression. This is better than human level, but it doesn't hurt to ask at this stage. Especially difficult is the notion of identifying a progression

scale of the illness. What we would like to do, that is the intelligence we would like to achieve, the superhuman intelligence we would like to achieve, is to diagnose Alzheimer's based on the description of subjects of the Cookie Theft scene. We aim to use speech because it contains emotional biomarkers that are lost with transcriptions. And there is a lot of evidence showing mood changes are early signs of Alzheimer's.

As you can see, all of our goals are a bit ambitious, and it positions our effort in the development side of the spectrum. This means that we won't find all the health solutions because everything that exists is below the targets we have set. We need to define the scope. It's the second important question of stage one. Here, we would like to focus on the onset of the disease. That is because there is a lot of evidence that the best place for medical intervention may be at the beginning of the disease. It is well-documented that brains of Alzheimer's patients end up distorted and much smaller in size.

The current hypotheses is that the neurological basis for this process starts early and takes the form of two toxic processes. The amyloid hypothesis is that one of these two processes generates amyloid plaques that become toxic. The second complementary hypothesis is that Tau tangles are accumulating, obstructing normal neural function. Causality is still unclear but there seems to be a consensus that both are part of the disease prognosis. The logic of our intentions is that if we could identify the beginning of these dual process and somehow stop it or delay it, the dementia impact would be drastically reduced because we would have tools to guide drug discovery in this critical moment of the disease.

Video 5: Stage II: Strategy and Operations — Alzheimer's and COVID (04:20)

At this point we have answered two questions related to artificial intelligence we want to focus on. The metric is 90% biomarker accuracy on disease discrimination and longitudinal progression, mimicking current tests. We did not specify if the accuracy referred to false positives or to false negatives because we'll have to analyze how the algorithms will be used before, we can decide exactly what we need. It's also important to take into account that we will be able to tweak the algorithms to tailor performance to the best possible use for our application. The second decision, scope, is the onset of the disease as defined by current medical best practices. Let's now focus on the two decisions for the second stage. Strategy and operations for the Alzheimer's and COVID case.

On the strategic side, focusing on Alzheimer's, of the three strategies, best product, network externalities, and full customer solutions, we'll focus on the first one, best product. Our goal will be to produce an AI pipeline that is best at AI diagnosis. This will mean that we will have the best audio diagnostics pipeline available for Alzheimer's. The reason to pick this strategy is that we expect zero variable costs, which means that if there is someone else with a better strategy, then they put us off the market by lowering costs. In many ways, audio diagnostics is sort of like the movie industry, where there is a winner takes all market.

You can see in the table shown on the screen, a list of failed clinical trials on Alzheimer's disease targeting one of the believed underlying causes of the disease, amyloid plaques. Thus, our best bet for this product responds to a need to help clinical trials targeting the onset of the Alzheimer's disease. We focus on this disease because it seems to be the one that has the most impact and attention, with the total global societal costs per year of the disease estimated at 1 trillion worldwide. To put this figure in context, there are less than 20 countries in the world with a GDP of this size or larger. The figure is larger than what you get if you add up the GDP of the bottom half of all the countries in the world. Thus, this product makes sense as a strategy. However, the other two strategies may also have some merit.

Let's review them just to view pros and cons. Full customer solution makes sense if you find an application where you can target AI discrimination efforts. For example, one may want to assess the effectiveness of a specific drug in a given cohort and our AD onset detector may help demonstrate drug improvements. Network externalities. They make sense here if one aims to have the largest database of Alzheimer's voice samples. There are a few research centers that are aiming to have this strategy. However, because of legal limitations, they can't really exploit their scale in the way they would like. Let's now move to COVID-19.

This last strategy that we just described for Alzheimer's, network externalities, may, in fact, make more sense for COVID-19 because of the time pressure. Given the uncertainties, it seems plausible that the first one to arrive to market can attract all AI diagnostic requirements and take advantage of the other two reinforcing components. Lock-in because probably it will be the only one legally approved for a while. And Scale because it will quickly get the most data and investment behind.

Video 6: Stage II: Operational Decision — Alzheimer's (02:17)

In terms of operations, we need to come up with at least one business process where the two will be usable. In the case of Alzheimer, we'll aim to support at least clinical trial development. The objective will be to develop biomarkers that can track progress between doctor assessments in the early phases of the illness. The value of doing so is to be able to help demonstrate whether a given drug helps stop the disease progression. Setting up clinical trials is a very long process, usually taking more than a decade and it requires several billion dollars in funding per drug. You can see on the screen the estimated clinical trial cost for an average Alzheimer's drug.

From an operational point of view, we'll have to engage a very large constituency to make sure that all the organizations involved, including the FDA, approve our algorithms as a medical device. This means that, from an operational point of view, we need to mobilize a clinical trial in health officials that can incorporate our AI diagnostic tool. We won't go into the details, but it's very well possible that the cost of putting our AI to work is on the order of 100 or more times larger than the cost of actually developing the AI itself. Introducing AI, in a clinical trial, will impact a lot of the tasks involved. Take any of the ones on the screen and you can see how AI needs to be taken into account already in project management.

Some Alzheimer's clinical trials involve hundreds of doctors and all will have to understand what is the role AI will play and have their say on the various possible scenarios. The cost of all these conversations are in the lower side of the iceberg. They are sometimes called collateral assets and many studies have demonstrated that IT projects fail because of lack of investments in these assets.

Video 7: IP Approach: The Choice of ResNets (05:09)

Most of the diagnostic applications in digital health use a machine learning approach like the ones reviewed over the last two modules. Here we will present an approach based on combining an architecture with a specific deep learning model. We will replicate the exact same model throughout the architecture because it makes it easier to conduct experiments and tinker with the overall architecture, which is an essential component of the next stage, stage four. We will describe the IP approach for Alzheimer and later briefly explain how it naturally extends to COVID. The fundamental unit, or model, we will be using are ResNets. It's important that you understand how easily it is to change this choice. Most likely, you will be using an AI programming packet like PyTorch or TensorFlow.

All of them make it easy to change from ResNet to other options like VGG or DenseNet. The piece of code on the screen lets you choose a model simply by changing the parameter of a call. Here the name is ResNet50, but everything that follows could be done with another option, such as

DenseNet or VGGs. That's why our IP strategy does not include creating any new type of deep learning network. Instead, what we're going to do is deploy these ResNet in a smart way, so that we can leverage the most existing programming platforms for our target task. ResNet50 is a good starting point for our ambitions, given we are focusing on sound, but in stage four, when we do tinkering, we'll make sure to try other options to see the results improve.

These are another example where there are a connections between the different stages. ResNet's are a good choice because they have been shown to improve regular convolutional networks. In the image you see a schematic of two convolutional networks on the left and one ResNet on the right. The key architectural innovation of ResNet is the introduction of the forward arrow. What this arrow does is signify the pushing the output of the layer of a few layers' upwards. On this other image, you can see a close-up of this basic architecture of block of a ResNet with this sort of arrow. This innovation was then to address a particular problem of convolutional networks that this term, the vanishing gradient problem. As you add layers to a simple convolutional network, the propagated gradient through backpropagation start diminishing to a point that it becomes practically random. This produces a phenomenon whereby adding layers actually decreases the performance of a system.

ResNets address this issue by making sure there is always some signal that's pushed forward so that the network is trained on residual differences. On this graph that you see on the screen taken from the paper that introduced ResNet, you can see on the left the results for convolutional networks of 18 and 34 layers. The 18-layer network outperforms the bigger 34-layer network consistently. On the right, use the ResNets where the deeper, larger layer network outperforms a shallow version. So, that the external computational investment has some benefits. So, only if we don't get satisfactory results will we develop a new type of network. This is what GPT3 did in fact. The investment there is so large, and they have so many resources to test things that they actually develop their own deep learning models. This is something we could do in later stages if we feel we can get benefit out of it, and we have the resources to do so. In general, even if you want to go that route, that is the route of new depth learning models.

My advice to you is that you first do a version with stand-alone solutions that can be used as a base case to compare with improved models you may develop on your own. Base case is a term I hope you retain for years to come as a designer of AI products. It's possibly going to be a very useful one for you when establishing comparisons between different generations of your products and those of your competition.

Video 8: Stage III — (Part 1: IP) The Choice of the Architecture (06:40)

Once we have decided for a basic tool, ResNets in our case, we could already move on and see what performance we can get on a discrimination task such as Alzheimer's. We did this and found we only got 60% accuracy. There are number of things you can do at this point. One of them is to use a different kind of model. Perhaps instead of ResNet50, try a larger network. We sort of already knew this because many people have tried and therefore, we had a more ambitious plan that I'll talk about next.

There are three things we introduced to improve these results. First, we created a graph with multiple ResNets and some other pre-processing that we'll describe next. Second, we did some data augmentation techniques to increase the size of our sample. And third, we used transfer learning. Let's now describe each of these three architectural improvements. I'll describe them for Alzheimer's and hopefully this gives you ideas on what kind of AI designs you can play within your own target tasks. And later, we'll talk about COVID which is a natural extension.

In our case, we designed an architecture inspired in MIT Center for Brain Mind and Machines Brain Model. Our hunch was that we could develop an Alzheimer's longitudinal discriminator by using different ResNets aiming to find biomarkers related to these four units of the brain. We call it the Open-Voice Brain Model Architecture. In your case, you may want to use a different inspirational reference model depending on whether you're working on a manufacturing plant or on some solar energy power grid. You can use any sort of model you have. The more subject expertise you have, the more likely you are to find a model that can help. The design principles we'll use can be applied to other domains as long as you have a reference model you can work with. In our case because Alzheimer's is a chronic disease of the brain, we postulated that the brain model would help and that was the main motivation for introducing the MIT Open-Voice Brain Model Architecture, or OVBM. The basic idea behind the architecture is to identify and intelligently combine biomarkers relating to each unit so that we could find malfunctioning components of the brain.

The CBMM four units of the brain are, the sensory stream is the first unit and is devoted with the first 200 milliseconds of perception in the human brain. It is responsible for taking input from sensors such as the eyes and ears and pre-processing the receiving information.

Alzheimer's/dementia impacts many aspects of a sensory stream including voice, hearing, and vision. The brain OS, the second unit, is able of taking input from the sensory stream and combining it with other modules to make progress. It's sort of like the unsupervised part of our brain that makes sense of all the inputs it receives with the cognition and stories it is building. The cognitive core is the third unit, and it's responsible for organizing areas of knowledge such as physical interactions, what you may know about plans or skills required for driving for example. Alzheimer's memory loss may be related to the cognitive core.

Finally, the fourth unit, the semantic compositional models are responsible for organizing stories. They may be able to perform the kinds of inferences that are required to explain the Cookie Theft drawing. They're also essential in having conversations or in writing a novel. Know that our choice of words is very specific to the Cookie Theft picture. It is very common in digital health to build biomarkers that are specific to a given task in a given sensor. This contrasts with the sensory stream ones that were more generic. You may find, in your applications, that you also have this duality, and your architecture may combine generic metrics with context specific ones. To understand how we have implemented the brain OS, please look at the architecture of the overall system.

Since we received as input a Cookie Theft free speech audio recording description that may be over a minute long, too long for a neural network, we chunk it into pieces of a few seconds long, six seconds in the diagram. There are a few techniques to deal with continuous signals such as videos or sound recordings. Chunking is a classic one, and there are several other options proposed, LSTM being a very popular one. For the COVID-19, we tried many architectures but what turned out to yield the best results was an architecture very similar to the one used for Alzheimer's. We did not use the complete brain model simply because we avoided the cognitive core biomarkers. For the rest, it is the exact same architecture as you can see.

One aspect that's important about where AI is going is the explainability component. Unfortunately, ResNet50s don't tell you a lot about what is going on inside them. On the other hand, if you use an architecture that has a certain logic, you may be able to offer some insights into how decisions are made. This concludes our overview of the first of the two decisions that need to be made in stage three of the process, the intellectual property decision or IP decision. Now we will move on to the data strategy decision.

Video 9: Stage III — (Part 2: Data Strategy) Privacy and IRB Approval (04:27)

Let's now describe the data strategy used for the Alzheimer's and COVID cases. In digital health, data is a big concern because of privacy constraints. That explains why there aren't many open data sets and why the use of data is highly constrained and closely guarded by data owners. In Europe, there is even a special regulation, very strict, called GDPR, which stands for general data protection regulation. We address privacy in the following way. In the case of Alzheimer's, we used a public data set called address that was "sanitized." Which means you don't have the risk of breaching any privacy issues. This, of course, is a moving target because one day, we may have a Shazam for voice that tells us who a voice corresponds to. We haven't reached this point yet. So, this is of no concern.

In the case of COVID, we used only coughs, which meant our recordings did not have private data. What is often called PII, or personally identifiable information, because it did, we were able to transfer it freely. We had some participating hospitals in Europe that would collect coughs and then send them to us along with verified PCR results that we could use to train our algorithms. In general, you need to go through an Institutional Review Board or IRB, which are specialized bodies within hospitals or universities before you can start collecting data about humans. Sometimes they are called ethics committees or COUHES, which stands for Committee on the Use of Humans as Experimental Subjects. And this applies if there's any sort of health data about the persons. Before you can collect health data, you need to describe what you want to collect and why to an IRB, and they have to approve it. In general, if one IRB approves it, you can use their approval to go to another one.

However, not all IRBs have the same criteria, and often companies will submit application to IRBs that may be favorable to what they are trying to do. IRBs look at the number of criteria such as scientific validity, privacy implications, risk, and benefits for participants, etc. Critical in the IRB process is the consent document. That is what people participating often call subjects will agree to. Here are two important pieces. A first piece in the consent outlines the purpose. In the COVID example we said purpose: in this study, we will collect sample coughs of COVID-19 patients and that of non-patients and try to distinguish the two groups using artificial intelligence. The application to the IRB, you also have to come up with a reason why participation is worth it for participants while outlining the possible risks and how to mitigate them. Sometimes participation is done for a fee. A second piece in the consent form is how data will be used. It's important here to have a use as generic as possible, including the possibility of commercial exploitation and use for other diseases and for other applications. If the second piece is not present, you may be restricted to what you can do with the data.

In terms of the collection strategy, the Alzheimer's case was pretty simple because we use databases that others had collected. This is a common strategy where you buy data from another source that gives you the rights you need. For COVID, we developed our own strategy, which I described.

Video 10: Stage IV — Tinkering — Software Methodology (05:03)

Let's now address the two decisions for stage four starting with a software methodology. They are very similar for COVID and Alzheimer's, so we'll only look at Alzheimer's and make some

comments on COVID where appropriate. Let's start with the software methodology. First, the code and data repository methodology often has different parts. In our case, the web and mobile app was done in React. It's a pretty simple piece of code, that does audio transfer and that is configurable. Here, the hardest part is the user interface. For the deep learning code, we use TensorFlow and GitHub. For our GPUs, we use the Azure cloud with NVIDIA Tesla machines. They are some of the most advanced in the market and can produce calculations equivalent to thousands of laptops. These GPU cloud architectures vastly augment what you can do. The more computation you have, the more training and experiments you can run.

Finally, we developed a testing and experimental plan. This is very important, especially if you plan to get your algorithms approved to become a medical instrument. In our case, we developed a plan to get past the first stage which is an academic publication. This then opens up the door so that others take your results and expand them via properly validated clinical trials for the specific purpose. In our case, we tested the OVBM architecture by selectively removing biomarkers using different configurations to find the best possible combination for both target diseases. In this figure, you see the percentage of Alzheimer's discrimination for various OVBM architecture deployments. All of them include the 16 biomarker models. In other words, they all have the same computational power and requirements. The one on the far left and field has no initialization for any of the biomarkers. We call this the base case and need to achieve a discrimination rate of 65%. As you move to the right, the solid bars indicate the full OVBM model with transfer learning. The bar with the zero indicates that transfer learning has only been done in the upper dense layer.

The next three adjacent bars to its right illustrate how the model improves when you let more layers change weights during transfer learning. The next three groups of bars represent the same architecture but where transfer learning is done with only one of the biomarkers. As you can see, we systematically added or removed a biomarker from a base case to be able to intelligently make decisions as to what brings value and what does not. I encourage you to do the same in your own tinkering plans. In our case, this exercise tells us, for example, that the biomarker them with zero transfer layers is as powerful as the base case or that all of the biomarkers greatly improve the base case with enough layer strength. One can also do some deep dives on specific topics. In this image, for example, we brought an experiment where we see what is the value of the biomarker cough by seeing how it improves both unpre-trained and pre-trained models. The message, as you can see, is that cough improves all biomarkers and therefore it is a net addition to the model.

In this other example, we show the percentage of subjects that are properly identified by each biomarker individually in the first three rows and in pairs in the following three rows. One can

observe that two of the biomarkers don't add anything to each individual one. We say that these biomarkers are orthogonal to each other because together they don't do more than the sum of them alone. The last row shows the subjects that are not properly classified by any biomarker alone and by the three combined. Know that the overall accuracy will always be better because the last deep layers of the network will combine all of the biomarker information. That's the features of all the biomarkers and none of the examples above assume the full network.

Video 11: Stage IV — Tinkering: AI Cancers (01:39)

Digital health has many AI concerns to address, as we discuss next. Explainability is one of the big issues because often diseases have multiple causes and symptoms, and the algorithms don't really tend to give a good explanation. In our case, we address this issue by simply generating a pre-screening tool to help doctors in providing individual information on the biomarkers. Another issue is that of bias. We have shown that biomarkers, like cough, are gender bias and even cultural bias. This means that we will have to do individual data gathering efforts for each culture. This is similar to what we already saw with language translation but with the additional complication that the biases are hard to identify, and there may be others.

For example, we found that our algorithms worked better in COVID for asymptomatic patients, which was a surprise for us. I hope that this overview of this digital health AI products has helped you develop a better understanding of what are the pros and cons of this technology. These cancers are often behind the unexpected rising costs of developing AI products when already in the tinkering stage.

Optional Video 1: Stage I: Figure-ground and AI Categorization — COVID Example (04:48)

With the Alzheimer's example, we've shown one could stop here and move on to stage two. However, for pedagogical reasons, we're going to extend the problem to a second condition. And we're going to imagine that we want to build a processing pipeline that can collect audio data and do health diagnostics of more than one disease. In particular, we're going to assume also that you want to diagnose in addition to Alzheimer's one other condition. We'll pick COVID-19 as an example, but one could imagine it's a different condition, or perhaps even a set of more than just one additional condition. Our aim is to design an intelligence where you cough into the phone and it tells you if it listens for a cough that's related to COVID-19 or not. The same sort of "intelligence" that tells you when you describe a picture you have dementia or not, you don't need the next devices to know that humans can distinguish some characteristics of coughs.

Listen, for example, to this cough. And now listen to this other cough. Both coughs come from a baby, and you can tell that fairly distinctively. If I speak with a nasal voice like I'm doing now you can probably tell me I'm doing so without looking at me. And you could do that almost for any human in the world. This example in a way is similar to the example of a car where you can tell if a window is open or not when you're inside a car on a highway. However, now listen to these three coughs. One has COVID-19 without symptoms. The other one has COVID with symptoms. And the other corresponds to a COVID asymptomatic. Listen to the first cough. Now listen to the second cough. And now listen to the third one. As far as we know, no human has been able to learn the skill of telling you which is which.

However, just like in the example of the air filters, AI could do it. We postulate that distinguishing COVID-19 coughs is possible and that with enough examples, we'll be able to distinguish between the different types of coughs with an accuracy of 99%, especially with asymptomatic people. There is a lot of research demonstrating that different types of coughs can be distinguished from each other using artificial intelligence. Experiments on Bronchitis, Bronchiolitis, Asthma, dry versus wet cough, have shown that this is possible. So, that will be our first decision in the case of COVID.

So, this 99% COVID-19 recognition, just like we did for Alzheimer's. In terms of scope in the COVID example, we'll want to focus again in the onset of the disease. Because we feel this is the best place to help the pandemic. If we could identify and isolate people with the virus at the beginning when they're just initially infected, in many ways we could solve the problem very quickly. Identifying absolutely everybody seems hard, but if we had a tool to diagnose at scale using the phone, groups of people, we will be much better equipped to manage the pandemic. Unfortunately, COVID-19 spreads silently through asymptomatic people, and that's why we put the emphasis on these type of subjects.

Optional Video 2: Operational Decision: COVID (Part 1 of 3) (06:16)

Let's now turn back to COVID. Here, the operational goal will have to be necessarily different. A longitudinal biomarker may be possible to develop but a lot harder to train because of the time window when AI may be relevant, one year, possibly two at most. One can make the case that an AI solution proven right for COVID may be appropriate for pandemics in the future so that what we develop can be sustainable in the long-term. Unfortunately, there aren't many examples of the use of AI in medicine. The lives of very few doctors and patients have significantly been affected by AI as of now. From an operational point of view, a COVID AI detector has the following advantages, which really extend to most other forms of AI use in medicine.

In other words, the advantages of a COVID cough detector over a biological testing from an operational point of view are at least the following ones.

First, you can run the test on a mobile device without the need to leave home. This means you don't put yourself nor others in risk. Second, you don't require training staff to take the test, drastically reducing the implementation costs in deployment. Third, you get the answer almost immediately, making it useful for applications that require immediate control, such as when going to restaurants, class, when traveling or when going to work. Fourth, you can do a test as many times as you wish. This means you can catch the virus as soon as it is feasible. You may have been exposed to a pathogenic organism and still be preclinically incubating the disease.

By repeating the test regularly, hopefully AI can pick up onset before you reach infectiousness even if you are asymptomatic. Fifth, the variable cost is practically negligible. This means great news for those with low purchasing power capabilities. Even if the test is not 100% accurate, perhaps one can develop a prescreening approach that greatly lowers the overall cost of testing a population sample no matter its size. This is where in the test can be done practically instantly from anywhere in the world by downloading it from the Internet or by running it on a web page. Seventh, every time someone collects the test, you can collect the cough data simultaneously and improve the algorithm. Here's where one can see network externalities at play.

Note how this contrasts with the use of Alzheimer for a drug clinical trial. There, the advantages are not as big. On the other hand, perhaps one can develop an Alzheimer-disease screening tool that does have disadvantages. I find that keeping these two examples in mind can help you contrast with whatever use of AI you want to implement. We'll cover three operational possibilities for COVID-19. The first, and one of the natural places to apply of COVID-19 discriminator, is to prescreen the whole population in a geography. In principle, the calculation required would allow for a whole population to be tested every day during the week. If the accuracy was 100%, then we can pretty much assume that governments will pick up the tool and the help we provide will be tremendous. Those that test positive may quarantine themselves for two weeks, and the spread of the virus will be reduced exponentially.

On the other hand, assume the algorithm is 95% accurate, meaning that it always makes a 5% error regardless of whether people have COVID or not. Then imagine a country with 10 million people, where the pandemics has just started and there are only 1,000 people infected. If we apply our COVID detector to the whole country, the algorithm would make a mistake in one half a million people and it say that almost all of them have COVID when only 1,000 really do. This is not a very desirable outcome. And on top of that, of the 1,000 people that are indeed affected, you would make the same 5% error and 50 people would go undetected, continuing to spread the virus. With a 90% accuracy we set as a target, the figures will be a bit better, but 10 people

would go undetected. And about 100,000 would be wrongly classified, creating all sorts of undesirable outcomes, possibly even panic. As a second example, then consider a system to filter who can enter a certain venue. For a restaurant or a company these figures may work because, at the microscale of an organization, if there is one outbreak in an area, it may help lower risk if other types of testing is not available.

The third example we'll introduce is poor testing, which is another option for our operational decision. Before I introduce it, let me cover a nuance about accuracy rates in AI that is very important, not just for medicine, but almost for any application, and something that you as an AI designer need to know.

Optional Video 3: Operational Decision: COVID (Part 2 of 3) (07:37)

Let's dig a bit deeper into what we mean when we set a target of an accuracy of 99% for our binary classification task. The intuitive notion is that we'll achieve a 90% accuracy is that you'll only make a mistake in 1% of the cases, regardless of whether you say that someone has COVID or not, or that you say that someone has Alzheimer's or not.

Unfortunately, this in general is very hard to ensure. Because in health, and in many other areas if not all of the other areas, discrimination tasks have different accuracies depending on whether the person has the disease or not. If someone has COVID, the algorithm may tell us 100% of the times that he has COVID. This will mean that there are no false negatives because it is impossible for the algorithm to tell us that the subject does not have COVID with does have it. The same algorithm, however, may get confused if the person does not have COVID and about 10% of the time tell us that he has COVID when he does not. This would mean a 10% false positive rate. On the screen, we've plotted how this algorithm performs in the four possible scenarios.

In this particular case, the overall accuracy may be 90% if we only test negative subjects; or 100% if we only test positive subjects; and 95% if we use what's called a balanced group of subjects to test the algorithm. In medical terms, we say that this algorithm has a specificity of 90%, and a sensitivity of 100%. AI algorithms, as you have learned, have different parameters, and we may be able to tweak the parameters so that we increase specificity. This is important, for example, in the case where you have very few infected COVID patients. If in a country of 10 million people you have 10,000 infected people, the algorithm would signal as positive these 10,000 plus 10% of the rest, which in this case would be close to a million. Meaning that your accuracy on the total population would be practically close to 90%.

The algorithms and the software then let you increase the specificity, but at the cost of decreasing the sensitivity. One can construct curves illustrating how the two parameters are

related, as in the chart you see on the screen which plots Alzheimer's discrimination based on PETs of the brain. Each curve is a different test. AUC (or area under the curve) is the proportion of the square that falls inside the curve. If the algorithm was perfect with 100% specificity and sensitivity, it would be one because it would be the whole area. Know that these curves differentiate well based on the sample population in terms of positives or negatives, but don't tell us anything about the onset of the disease. This means that if you do your tests on a population of very advanced Alzheimer's patients, chances are your AUC will be much better than if you test at onset. Here, we come to a second very interesting discussion on metrics. What does onset mean in terms of health diagnostics? The reality is that there isn't a well-established answer.

Recent research has shown that there may be chemical traces of the beginning of Alzheimer's up to a few decades before the clinical onset of the disease. However, a person that seems destined to have the disease may not develop it because of several reasons. There is some evidence that one can stop the progression of the disease if intervening early. There is also evidence that behavioral elements have an effect on the speed of development. All of this confounds with the definition of onset and makes it really difficult. The same thing happens with the COVID virus, where it's unclear exactly when does the disease start. It may be that you are a carrier of the virus and not develop the disease for a while, as it happens with multiple virus. And if you develop it, it may take a while before it is noticeable, making it impossible to detect the virus or not.

In some cases, the AI algorithm may detect the disease, but you have incorrectly labeled data, because you had no way to know the ground rules. You may think that the AI is wrong when, in fact, it is right. This is the case with COVID-19 because the chemical tests are not perfect. And also with Alzheimer's, because sometimes not even the doctors agree on whether someone has Alzheimer's or not. That's life in the big city for AI designers. In our case, we will define onset as the earliest any other method can detect the presence of the disease. What we want our AI intelligence algorithm to do is to give us some metrics about the progression. These metrics, when referring to biological systems, are often called biomarkers. Alzheimer's disease is a disease that gradually progresses. And we are interested in having a metric of how advanced the disease is.

Or, in other words, we want the algorithm to give us biometric estimates that indicate how serious the condition is. Temperature, for example, is one such biomarker. It gives us some indication of how a subject is feeling, and if above a threshold, can signal life-threatening conditions. The term that is often used is longitudinal. Meaning that you have a medical biomarker that evolves and gives you an idea of the disease progression. That's why people call

these metrics longitudinal biomarkers. With COVID, one can also imagine various longitudinal biomarkers that tell us how seriously affected a subject is. As we will see later, the exact accuracy we need in terms of these biomarkers and how they relate to false positives and false negatives will depend on the medical use. In many ways, the eight decisions are interrelated. And that is why the role of the AI designer is so important.

Optional Video 4: Operational Decision — COVID (Part 3 of 3) (07:00)

As we have seen in the previous videos, AI has a knob that allows you to play a bit with thresholds to influence how many false positive and false negatives, the discrimination algorithm will produce. So, you can optimize your operational approach. Our third operational proposal will take advantage of this feature, you'll see how. The idea is to apply the test to groups to see if there is a COVID outbreak within the group. The intuition behind the approach is simple. If you have an algorithm that gives you 5% false positives and 5% false negatives and you apply it to groups of 100 people that have zero cases, you expect to have 5% false positives. Know that false positives are irrelevant in this case because you assume there is no single positive case. This means that on average you'll get 5% false positives or about 5 people in the group.

As long as the number is around 5, you are quite confident that the assumption can be right. Now, assume that in a group you get 50 positives, and the group has only 100 people. It sounds unlikely that those 50 can all come from false positives. Since the AI algorithm is like tossing a coin that 5 out of 100 times gives the wrong answer, it is possible. To have these 50 positives, you would need to toss a coin that 95% of the time gives you A instead of B and get 50 Bs with only 100 throws. Now, how likely is this numerically? Think about it. We can actually calculate this number. Those of you that know statistics probably have already calculated it, and it's an insanely low number. I mean, in fact, it's 100 factorial over 2 times 50 factorial. This is as likely as randomly picking up two atoms of the whole Earth and finding exactly the same. In other words, if you test the whole world and in one group of 100 people there are 50 that are positive, you can be 99.9999999%, with as many 9's really as you want, that in this group there are COVID positives.

One can make this calculation for any rate of false positives and false negatives and for any group size. Since we don't know exactly how good the AI algorithm will be, as long as we're still in Stage two, let's make some calculations to see what the impact is of AI improvements. In the diagram you're seeing, we have plotted several possibilities for groups of 25 people based on the individual accuracy of the algorithm. Here we assume that it's symmetric in terms of false positives and false negatives. On the x-axis, we picked the accuracy of the algorithm we developed, starting with 88% and ending with 100%. This is the algorithm applied to just one

person. We're taking an algorithm that discriminates between positives and negatives with a given accuracy as shown in this x-axis. On the y-axis, we plot the number of cases that are required for a desired group accuracy. That is, the number of cases, in this case of a group of 25, that must have COVID so that we can detect it for the whole group. And we plot three curves with three possible group accuracies 99, 95%, 99%, and 99.9% accuracy.

Let's see what the point in the curve means. Take, for example, the middle point on the blue line, and it represents that to have a 99% certainty that the group has a positive case of COVID-19, if our AI can detect it individually with 97% accuracy, then we need four cases. In other words, our AI, for groups would correctly label with a 99% accuracy, groups of 25 people, as having COVID if at least four people in the group have it. If you wanted the accuracy to increase to 99.9%, you only need to increase the number of cases by one more, that's up to 5. Let's look at an example where we only achieve 90% accuracy. And let's assume we want to reach a 99.9% accuracy for groups of 25 people, then the graph tells us that we need nine people.

In a group of 25, this is about 1/3 of the people. Let me finish this review of Stage two with three thoughts. First, once you finish Stage two, it is a good moment to pause and think about the use cases that one wants to support to make sure the right objectives are in place for Stage three when key technical decisions will have to be made. For example, depending on the use case you intend to implement, you may need a higher or a lower accuracy. And this means you may in turn require more or less data for training, which is key when embarking on Stage three and four. Second thought, the discussion on Stage two should also highlight the importance of the AI designer in the whole process. And, finally, third, as you can see in the first round, it's okay to keep a few operational options open until you find out what your AI can really deliver. In fact, it's good to keep your options open until you come up with something that really works.

Optional Video 5: Stage III — COVID Data Acquisition Strategy (03:08)

If we succeed in our COVID-19 prescreening tool, the impact could be enormous. And it is very possible that the phone app has so many users that collecting data will be automatic. One can even imagine a scenario where smart speakers, phones, and web browsers all collect samples as they produce screening results, making the database even bigger. This would mean a data strategy that has network externalities as we discussed in the first module of the course. Here we will be producing the virtual cycle, where the more people download the app, the more cough recordings we have. And, therefore, the more economies of scale we'll have and the more system lock-in we produce. Here system lock-in can be achieved in several ways, including by approved IRB protocols that everyone uses by the download of the apps, by the longitudinal data capture from single users, and by companies setting up pool testing.

One of the key difficulties in getting this cycle started is getting enough data at the beginning to get the ball rolling. In our case we created a crowd-sourced effort and engaged the media to help us get the word out. In general, it is very difficult to obtain spontaneous participation, and we found that the average media exposure value per cough recording was about \$100. In general, only 1 to 3 per 1,000 people did contribute their cough. The advantages of crowd sourcing data in a pandemic like COVID-19 come on top of the seven already mentioned advantages of using a mobile app for prescreening.

Here are some of these additional advantages. Eight, people can volunteer their data from the home without the need to go to a dedicated lab facility. All of our crowd-sourced coughs, in fact, came from recordings than outside medical facility. Nine, there is really no need for trained personnel to collect the samples. Ten, data gathering can be done instantaneously and at scale. Eleven, the variable costs for subjects and the AI algorithm are practically zero. Twelve, given the above, one can conceivably collect longitudinal data from the same subject, which is practically nonexistent in other forms of data collection. For validation we also initiated a second data gathering effort, consisting in hospitals from around the globe receiving cough recordings via WhatsApp. The hospitals would then cross-reference recordings received with official PCRs and send a sanitized version of this. This means that phone numbers was taken out from the recordings alongside any possible PII.